

Executive Summary — Volt-Bike Operational Performance Analysis

Business Objective

The goal of this analysis was to test one main assumption:

improving production processes and spending more on operations would increase customer satisfaction.

To evaluate this, the project used:

- KPI analysis
- correlation analysis
- variability tracking

to understand whether operational improvement really creates business value.

Key Findings

- The average assembly time is around 60 minutes, but only about 16% of bikes are assembled within $\pm 5\%$ of that average. This shows that some variability still exists in the process, although the production system is already fairly mature.
- The relationship between production cost and customer satisfaction is weak to moderate ($R^2 \approx 0.23$). This means that spending more money on production does not always lead to higher customer satisfaction.
- Customer satisfaction stays close to 6.5 for all bike types. This suggests that the products are not very different from each other, and the company may need more innovation to stand out in the market.
- The relationship between assembly time and customer satisfaction is extremely weak ($R^2 \approx 0.0003$). This means production efficiency has almost no effect on how customers feel about the product.

Executive Insight

- The analysis suggests that the production process is already working at a stable level.
- Further optimization may increase costs without creating meaningful improvements in customer satisfaction.
- Instead of focusing mainly on process improvement, the company should focus more on product innovation and customer value. "Don't build it better — build it differently."

Recommended Strategic Actions

- Invest more in product innovation and new features. Priority: HIGH
- Conduct deeper customer research to better understand customer needs and expectations. Priority: HIGH
- Keep the current production standards and improve operations only when necessary. Priority: MEDIUM
- Avoid large process optimization projects unless they clearly support innovation or customer value. Priority: LOW

Executive Takeaway

- The current production process is already stable, and more optimization is unlikely to improve customer satisfaction in a major way.
- Customer satisfaction depends more on what the company builds than on small improvements in production precision.
- Long-term competitive advantage will come from innovation, not from small operational improvements.
- "Efficiency supports the business — innovation makes the business different."